CSIT321 PROJECT DESCRIPTIONS
JULY 2026 – DECEMBER 2026

Projects available:

CSIT-26-S3-01	**Privacy-Preserving AI Assistant for Smart Campuses**
CSIT-26-S3-02	**Secure QR-Based Access and Identity Verification System**
CSIT-26-S3-03	**Explainable AI Platform for Health and Lifestyle Recommendations**
CSIT-26-S3-04	**AI-Driven Detection of Malicious Smart Contracts**
CSIT-26-S3-05	**Federated Learning System for Mobile Image Recognition**
CSIT-26-S3-06	**Intelligent Scheduling and Allocation System with Fairness Constraints**
CSIT-26-S3-07	**Human-in-the-Loop Intrusion Detection Dashboard**
CSIT-26-S3-08	**Secure Data Sharing Platform for Collaborative Research**
CSIT-26-S3-09	**Trust-Aware Recommendation System for Online Communities**
CSIT-26-S3-10	**AI-Based Misinformation Detection for Short-Form Content**
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CSIT-26-S3-21	**News Release System Design and Implementation**
CSIT-26-S3-22	**Dog Identification System**

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Project ID	CSIT-26-S3-01
Proposed Title	**Privacy-Preserving AI Assistant for Smart Campuses**
Project Description	This project aims to design an AI assistant that supports campus services such as room booking, event reminders, and navigation, while ensuring user privacy through local processing and cryptographic techniques. Students will explore privacy-preserving machine learning, lightweight encryption, and secure data storage to ensure that personal usage data is never exposed to central servers.

Project ID	CSIT-26-S3-02
Proposed Title	**Secure QR-Based Access and Identity Verification System**
Project Description	Building on existing QR security and authentication ideas, this project focuses on designing a secure QR-based identity and access control system for facilities such as libraries or labs. The system should integrate biometric verification, replay-attack protection, and QR payload validation, with a formal security analysis of potential threats.

Project ID	CSIT-26-S3-03
Proposed Title	**Explainable AI Platform for Health and Lifestyle Recommendations**
Project Description	This project develops a recommendation system for health or fitness that not only produces personalised suggestions but also explains its reasoning in human-understandable terms. The focus is on explainable AI techniques, fairness, and user trust, rather than purely optimising prediction accuracy.

Project ID	CSIT-26-S3-04
Proposed Title	**AI-Driven Detection of Malicious Smart Contracts**
Project Description	This project investigates how machine learning and static analysis can be used to detect vulnerabilities and malicious patterns in smart contracts. The system should analyse contract code, identify risky behaviours, and provide interpretable warnings. This project blends blockchain security with program analysis.

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Project ID	CSIT-26-S3-05
Proposed Title	**Federated Learning System for Mobile Image Recognition**
Project Description	Students will design a federated learning framework where image recognition models are trained collaboratively across mobile devices without sharing raw data. The project explores model aggregation, communication efficiency, and privacy leakage risks, with a practical prototype on simulated clients.

Project ID	CSIT-26-S3-06
Proposed Title	**Intelligent Scheduling and Allocation System with Fairness Constraints**
Project Description	Extending beyond basic optimisation, this project focuses on designing a scheduling or allocation engine that explicitly models fairness, workload balance, and stakeholder preferences. Possible applications include project allocation, task assignment, or volunteer rostering, with transparent decision logic.

Project ID	CSIT-26-S3-07
Proposed Title	**Human-in-the-Loop Intrusion Detection Dashboard**
Project Description	This project develops a next-generation intrusion detection interface that combines automated alerts with human feedback. Students design adaptive visualisations that learn from administrator responses to reduce false positives and improve operational usability.

Project ID	CSIT-26-S3-08
Proposed Title	**Secure Data Sharing Platform for Collaborative Research**
Project Description	The project aims to build a system that allows researchers to share sensitive datasets while enforcing fine-grained access control, auditing, and revocation. Techniques may include attribute-based encryption, access policies, and secure logging, with a clear real-world research use case.

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Project ID	CSIT-26-S3-09
Proposed Title	**Trust-Aware Recommendation System for Online Communities**
Project Description	This project focuses on designing recommendation algorithms that incorporate trust, reputation, and risk signals rather than only user similarity. The system should explain why content or users are recommended and analyse how trust evolves over time.

Project ID	CSIT-26-S3-10
Proposed Title	**AI-Based Misinformation Detection for Short-Form Content**
Project Description	Students will design a system to detect misleading or manipulated information in short-form posts, images, or captions. The project highlights data quality challenges, model robustness, and ethical considerations, with an emphasis on explainability rather than content moderation.

Project ID	CSIT-26-S3-11
Proposed Title	**Lightweight Cryptographic Toolkit for Mobile and IoT Applications**
Project Description	This project involves building a practical cryptographic toolkit optimised for constrained environments. Students will implement, benchmark, and compare lightweight encryption, authentication, and key management schemes suitable for mobile or IoT systems.

Project ID	CSIT-26-S3-12
Proposed Title	**Emotion-Aware but Privacy-Respecting Text Analysis System**
Project Description	Unlike sentiment analysis platforms that centralise user data, this project explores how emotional cues in text can be analysed locally or with minimal disclosure. The system should demonstrate how insights can be extracted without reconstructing raw messages.

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Project ID	**CSIT-26-S3-13**
Proposed Title	**Go Climb Mobile Application**
Project Description	This is a web based application for outdoor sports such as Rock climbing primarily. This app will let the users search for crags in Australia (or a place of your choice, get details on each crag, potential hazard based on the day's weather and Augmented Reality integration for image and route tracing. The app will also let users create profile which will display all their climbs and achievements on the user dashboard and connect the climber community through chat forum. Back end: Python Front end: Django and Javascript Unity and vuforia for AR

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Project ID	CSIT-26-S3-14
Proposed Title	**Biometric Authentication Security Mobile Tool**
Project Description	This project examines the security of biometric authentication (e.g., fingerprint, facial recognition) in mobile apps. It assesses vulnerabilities, particularly side-channel attacks, and proposes strategies for enhanced security. The tool collects biometric data, conducts statistical analysis, and identifies potential weaknesses. It offers mitigation suggestions, such as altering data transmission protocols or countering timing-based attacks. The user-friendly interface lets users select methods, analyse vulnerabilities, and access recommendations. The project includes the following component. 1. **Biometric Data Collection Module:** Develop modules to collect biometric data (fingerprint, facial features, etc.) from mobile devices' sensors. Ensure the data collection process is accurate and follows privacy regulations. 2. **Vulnerability Assessment:** Analyse the collected biometric data to identify potential vulnerabilities. Investigate side-channel attacks that might exploit the biometric data's transmission or processing. 3. **Statistical Analysis and Correlation:** Create algorithms to statistically analyse the biometric data and establish baselines. Detect variations in the data that could be indicative of vulnerabilities. 4. **Recommendation Engine:** Develop a recommendation engine that suggests actions to enhance the security of biometric authentication methods based on the analysis. This could involve proposing changes to data transmission protocols or processing algorithms. 5. **Mitigation Strategies:** Design strategies to mitigate vulnerabilities and potential side-channel attacks. Develop techniques to counteract timing-based attacks or other types of attacks that exploit biometric data leakage. 6. **User Interface:** Create an intuitive user interface for the tool. Users should be able to select biometric methods, initiate analyses, view vulnerabilities, and access mitigation recommendations. 7. **Integration and Testing:** Integrate the various modules into a cohesive tool. Thoroughly test the tool using different biometric authentication methods and scenarios to validate its effectiveness. 8. **Documentation and Presentation:** Prepare comprehensive documentation detailing the project's design, implementation, testing procedures, and results. Craft a presentation that effectively communicates the objectives, methodologies, and outcomes of the project

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Project ID	CSIT-26-S3-15
Proposed Title	Secure Online Payment Gateway Design
Project Description	This capstone project revolves around crafting a highly secure online payment gateway, safeguarding financial transaction confidentiality and integrity. It entails implementing encryption protocols, user authentication mechanisms, and fraud detection systems to ensure secure transactions within the digital landscape. The project includes the following components. 1. **Encryption and Data Security:** The project involves developing and implementing encryption protocols to protect sensitive payment data during transmission. Adherence to security standards is paramount to prevent data breaches. 2. **User Authentication Framework:** Creating a robust user authentication system for customers and merchants is a core aspect. Multi-factor authentication will enhance security by verifying user identities effectively. 3. **Transaction Integrity Assurance:** Ensuring the legitimacy of transactions is crucial. Designing measures like digital signatures or checksums will help detect any unauthorised tampering attempts. 4. **Fraud Detection and Prevention:** The project includes designing algorithms for spotting suspicious activities and transaction anomalies. Automating processes to flag or halt potentially fraudulent transactions is a significant focus. 5. **User-Friendly Interface:** Balancing stringent security measures with user-friendly design is challenging. The project emphasises an intuitive interface that maintains security without compromising usability. 6. **Integration and Testing:** Integrating the project's components into a cohesive, secure payment gateway system is pivotal. Rigorous testing across scenarios ensures robust security and performance. 7. **Documentation and Presentation:** Comprehensive documentation detailing project blueprints, strategies, testing methods, and results is essential. A concise presentation encapsulating accomplishments concludes the capstone.

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Project ID	CSIT-26-S3-16
Proposed Title	**Interactive Website for Real Estate Price Prediction in Singapore**
Project Description	The project involves designing a user-friendly platform to predict real estate prices in Singapore using machine learning. The project encompasses data collection, pre-processing, and feature engineering to train accurate predictive models. Users can interact with the website, input property details, and receive instant price estimates based on historical data and property attributes. The project includes the following components. 1. **Data Gathering and Processing:** Collect comprehensive real estate data, including property characteristics, transaction history, and location-based features. Clean and pre-process the data for analysis. 2. **Model Development:** Select suitable machine learning algorithms, train predictive models, and fine-tune them using historical data. Evaluate model performance using appropriate metrics. 3. **User Interface Design:** Create an intuitive web interface where users can input property information. Design an engaging and responsive user experience. 4. **Geospatial Insights:** Incorporate geospatial analysis to account for location-related factors. Assess the impact of amenities, transportation, and neighbourhood attributes on property prices. 5. **Data Visualization:** Generate visualisations to showcase predicted prices, trends, and geographical influences - present insights in an easily understandable manner. 6. **Deployment and Accessibility:** Deploy the trained model on the website, enabling users to obtain instant price predictions for their properties. Ensure accessibility across devices.

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Project ID	CSIT-26-S3-17
Proposed Title	**Lightweight Image Recognition App for Singapore Botanic Gardens**
Project Description	The goal of this project is to develop an image recognition app to provide information about specific plants, flowers, and architectures in the Singapore Botanic Gardens. This app, which can be installed in a smartphone, can enhance the visitors' experience when visiting the gardens. It can help visitors discover new interests, learn new things, and have fun while doing so. The app should include the following key features: • Image classification capabilities using pre-trained Artificial Neural Networks (ANN), such as MobileNetV3, to classify the plants and architectures. • Integration with the smart phone's camera. • Integration with an Internet search engine, which allows users to learn more information about the recognized object from Internet. • Integration with a popular social platform (e.g., Facebook Messenger, WhatsApp).

Project ID	CSIT-26-S3-18
Proposed Title	**Chatbot App for Book and Video Recommendations for Kids**
Project Description	The goal of this project is to develop a chatbot app to recommend books and videos for kids. The chatbot will be able to respond to queries about age-appropriate books and videos for children, as well as provide additional information about the content. The app should include the following key features: • Natural Language Processing (NLP) capabilities to understand and respond to user queries. • A database of books and videos for kids, including information about the content, age appropriateness, and other relevant features. • Recommendation engine that leverages the ANN models and database to provide personalized recommendations based on user queries and preferences. User interface that is user-friendly and easy to navigate, allowing users to interact with the chatbot and browse recommendations.

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Project ID	CSIT-26-S3-19
Proposed Title	**Developing recommendation app for diet and nutrition**
Project Description	The aim of the project is to design and develop a mobile application for diet and nutrition. The students should conduct a market and literature research in order to explore different types of recommendation apps, as well as the features offered by the recommendation apps/systems in nutrition and dietetics. Based on these analyses they need to design and develop a mobile application that can be used for dietary recommendation. The aim of the app is to help users to develop a healthy diet and track their progress using the app. Some of the main functionalities should include: o allowing user to define the goal (calorie limit per day), o generate meal recommendations based on the user's calorie goal, o allowing user to insert the calorie intake per meal, o using online API to obtain recipes for the recommendations, o allowing user to insert recipes, o user diary, daily/monthly calorie intake (diet) reports. *Including advanced feature like recognising meal (calculating calories) from image (taken by the user) can be an excellent addition to the app.

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Project ID	CSIT-26-S3-20
Proposed Title	**An Online Tournament Management Platform**
Project Description	This project aims to develop a mobile application (for Android, iOS, or both) that can collect exercise data from cellphone sensors or wearable devices, provide training and fitness advice, and facilitate social interactions such as initiating competitions between registered users. The application should have the following features. 1. Collect exercise data: The application can collect user exercise information from either the cellphone's own sensors or from smart wearable devices (via Wi-Fi, Bluetooth, etc.). 2. Estimate exercise effects (short-term or long-term) and provide analysis: The application can estimate the exercise results of each activity, a day, a week, a month, or any period, such as the number of calories burned and the average and maximum heart rate. 3. Supply fitness advice and schedule a fitness plan: The application can provide fitness advice according to the user's fitness goals and their entered health information. It can also create a fitness plan based on the fitness goal, and users can customize the plan to their preferences, such as the length of the plan or the daily weight loss goal. 4. Remind user of exercise or take a break: The application can send reminders to users to exercise if they have not met their scheduled exercise goals. It can also send reminders to users to take a break if they have been exercising too much. 5. Connect with social media and initiate competitions between users: The application can be used to share exercise completion data on social media platforms such as Facebook, Instagram, and Twitter. Users can also initiate competitions with other registered users and invite them to join in. The competition results and rankings can be shared on social media. The project may consider releasing the application to the app store.

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Project ID	CSIT-26-S3-21
Proposed Title	News Release System Design and Implementation
Project Description	Today's news release system is different from traditional media. Traditional media mainly focus on dissemination of news information, while modern news release systems not only offer users the ability to read and browse news content, but also allow users to express their views and to communicate with other users. The system function can be divided into two main components, which are a management part and a display part. The news release system should allow the non-registered users to read news articles. But for non-registered users, they will not be able to comment on the news. If a registered user logs in, they will be allowed to comment on news articles, publish their own news pieces, and share the updates with other users in real-time. Additionally, ordinary registered users can manage their personal news articles. These articles can also be subject to management by the administrator, which ensures the security and standardisation of the system.

Project ID	CSIT-26-S3-22
Proposed Title	Dog identification system
Project Description	The idea is to identify via smart phone, what kind of dog that stands in front of us. For example, the app will say this is a border collie. Then, the app should be able to identify if there is any issue with the dog, such as any disease to come. This project should make use of AI driven approach. It should link with the AI capabilities that can help identifying issues.

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